

Hierarchical Reinforcement Learning for Adaptive Large Neighborhood Search in the Vehicle Routing Problem with Time Windows

HUYNH NHAT HUY¹, AND HO THI LINH²

¹Faculty of Information Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam (e-mail: huynhnhat@tdtu.edu.vn)

²Natural Language Processing and Knowledge Discovery Research Group, Faculty of Information Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam (e-mail: hotlinh@tdtu.edu.vn)

Corresponding author: Thi-Linh Ho (e-mail: hotlinh@tdtu.edu.vn).

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ABSTRACT The Vehicle Routing Problem with Time Windows (VRPTW) is a fundamental combinatorial optimization problem in logistics, characterized by a hierarchical objective that prioritizes minimizing the total number of deployed vehicles (N_V) before minimizing total travel distance (D_T). While classical Adaptive Large Neighborhood Search (ALNS) remains a widely used and competitive metaheuristic paradigm for vehicle routing, baseline implementations relying on heuristic score aggregation and static acceptance criteria can encounter search stagnation in complex spatial-temporal topologies. Conversely, pure end-to-end Deep Reinforcement Learning (DRL) methods face feasibility challenges under hard time windows and out-of-distribution generalization barriers. To bridge these paradigms, this paper presents Tri-Level Hybrid DDQN-ALNS, a learning-augmented optimization framework structured into three coordinated decision levels: (i) a Macro Decision Tier (Plateau Controller) formulated as a semi-Markov Decision Process to monitor search stagnation and govern regime transitions, (ii) a Micro Selection Tier where a Dueling Double Deep Q-Network (DDQN) with an explicit normalized Softmax-Entropy confidence gate learns state-conditioned destroy-repair pairings, and (iii) a Learned Acceptance Criterion (LAC) that evaluates downstream solution promise over a lookahead horizon. The low-level search engine leverages an $O(1)$ incremental forward-time-slack evaluator per candidate insertion and spatiotemporally pruned granular neighborhoods, combined with periodic exact Set Partitioning recombination over elite routes. We evaluate the proposed framework across standard Solomon-100, Homberger-200, and Homberger-400 benchmarks under strictly isolated cold-start execution protocols, assessing fleet-size stability, routing efficiency under matched fleet counts, and computational trade-offs.

INDEX TERMS Adaptive Large Neighborhood Search, Deep Reinforcement Learning, Double DQN, Hierarchical MDP, Metaheuristics, Vehicle Routing Problem with Time Windows.

I. INTRODUCTION

The Vehicle Routing Problem with Time Windows (VRPTW) is a foundational NP-hard combinatorial optimization problem underpinning modern freight logistics, last-mile parcel delivery, and sustainable distribution networks [?], [?]. In addition to the combinatorial explosion inherent in classical routing, VRPTW enforces hard customer time-window constraints: each vehicle route must simultaneously respect vehicle capacity limits and temporal feasibility at every stop.

In commercial freight operations, the optimization objective is strictly lexicographic: dispatchers prioritize minimizing the total number of deployed vehicles (N_V) before minimizing the total travel distance (D_T) [?], [?]. Formally, letting $\mathcal{S}$ denote the space of feasible routing solutions, a solution $s_1 \in \mathcal{S}$ is strictly preferred over $s_2 \in \mathcal{S}$ (denoted $s_1 \prec s_2$) if and only if:

$$NV(s_1) < NV(s_2) \vee (NV(s_1) = NV(s_2) \wedge TD(s_1) < TD(s_2)) \quad (1)$$

This hierarchical ordering reflects the economic structure of commercial transportation, where fixed vehicle capital acquisition, maintenance, insurance, and driver shift wages dominate marginal fuel consumption.

Adaptive Large Neighborhood Search (ALNS) [?], [?] has long served as a leading metaheuristic paradigm for VRPTW, exploring diverse neighborhood structures via iterative ruin-and-recreate (destroy and repair) operations while strictly preserving feasibility. However, standard ALNS implementations rely on a stateless roulette-wheel or multi-armed bandit mechanism that updates selection probabilities from aggregated historical reward averages. This design exhibits two structural weaknesses:

- 1) **Stateless Operator Selection:** Heuristic selection weights depend solely on historical scores across coarse iteration segments, ignoring the instantaneous search state, such as spatial route fragmentation, remaining temporal slack, or capacity utilization.
- 2) **Myopic Plateau Stagnation:** Classical selection mechanisms struggle to coordinate multi-step operator sequences required to escape structural plateaus, frequently stalling when fleet-size reduction requires evaluating temporarily non-improving intermediate states.

In parallel, Neural Combinatorial Optimization (NCO) has emerged as a data-driven alternative, using Deep Reinforcement Learning (DRL) models such as Pointer Networks [?], Attention Models [?], and reinforcement learning frameworks for routing [?]. While these approaches capture spatial regularities without manual heuristic engineering, direct application of pure end-to-end neural solvers to industrial VRPTW encounters significant challenges:

- 1) **Feasibility Degradation under Tight Time Windows:** Autoregressive decoding policies often face constraint conflicts when customer time windows are narrow, leading to high constraint violation rates or requiring surrogate penalty tunings [?], [?].
- 2) **Objective Misalignment:** Most neural formulations optimize scalarized travel costs rather than strictly enforcing the primary-secondary fleet minimization hierarchy ($N_V \succ D_T$).
- 3) **Out-of-Distribution Sensitivity:** Pure constructive policies trained on synthetic uniform instances often experience performance degradation when applied to larger instances or heterogeneous spatial topologies without extensive fine-tuning.

To bridge these paradigms, this paper presents Tri-Level Hybrid DDQN-ALNS, a learning-augmented optimization framework that embeds state-conditioned reinforcement learning directly inside the iterative search loop of ALNS. Rather than replacing the metaheuristic with an opaque black-box neural solver, our approach maintains hard operational constraint feasibility and interpretable neighborhood operators from operations research, while replacing stateless roulette-wheel selection with a multi-level reinforcement learning control architecture.

The primary contributions of this paper are summarized as follows:

- **Hierarchical Control Architecture:** We formulate metaheuristic search management as a Hierarchical Markov Decision Process (HMDP), decoupling macro-level search exploration regimes (Plateau Controller) from micro-level destroy-repair operator selection (Operator Controller).
- **State-Conditioned Micro Selection with Entropy Gating:** We develop a Dueling Double Deep Q-Network (DDQN) operating over a 20-dimensional spatiotemporal state space, incorporating a normalized Softmax-Entropy confidence gate to adaptively balance learned neural predictions with Dirichlet/Thompson sampling priors.
- **Learned Acceptance Criterion (LAC):** We introduce a supervised neural acceptance model trained on delayed downstream lookahead signals, replacing static Simulated Annealing cooling schedules with state-aware evaluation of long-term solution potential.
- **Rigorous Scale-Aware Benchmark across 74 Instances:** We evaluate the framework across 74 standard benchmark instances (56 Solomon-100, 12 Homberger-200, and 6 Homberger-400 instances) under strictly isolated independent cold-starts, providing an empirical characterization of solution quality, fleet reduction, and computational trade-offs supported by non-parametric statistical hypothesis testing.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III presents the formal problem formulation. Section IV describes the proposed Tri-Level Hybrid DDQN-ALNS methodology. Section V details the empirical benchmark results, statistical tests, and ablation analyses. Section VI discusses scale-aware dynamics, analytical properties, and limitations. Finally, Section VII concludes the paper and outlines future directions.

II. RELATED WORK

A. CLASSICAL METAHEURISTICS AND ALNS FOR VRPTW

Metaheuristics have long dominated the practical resolution of the VRPTW [?]. Building on the seminal insertion heuristics of Solomon [?] and the Large Neighborhood Search (LNS) paradigm of Shaw [?], Ropke and Pisinger [?], [?], [?] formalized the Adaptive Large Neighborhood Search (ALNS) framework, in which a portfolio of handcrafted removal and insertion operators competes for selection via roulette-wheel probabilities updated from historical performance scores.

Subsequent research advanced two complementary directions. For the hierarchical objective of fleet minimization, Guided Ejection Search [?] and penalty-based edge assembly memetic algorithms [?] remain the strongest dedicated mechanisms. In parallel, matheuristic hybrids exploit an elite route pool—tracing back to the adaptive memory of Rochat and



TABLE 1. Taxonomy of Representative Methodologies for VRPTW and Related Routing Problems

Methodology Category	Representative Works	Hard TW Feasible	Objective Paradigm	State-Aware Selection	Hierarchical Control	Learned Accept
Classical Metaheuristics	ALNS [?], [?]	✓	Lexicographic	× (Stateless)	×	× (SA)
	SISR [?], HGS [?]	✓	Lexicographic	× (Rule-based)	×	× (SA/Threshold)
End-to-End Neural Solvers	AM [?], DRL-VRP [?]	× (Penalized)	Scalarized	N/A	×	N/A
	Neural VRPTW [?], [?]	× (Mask-fragile)	Scalarized	N/A	×	N/A
Single-Agent Hybrid RL-LNS	Learned LNS [?], [?]	✓	Scalarized	✓	× (Single-agent)	× (SA)
	Neural LNS [?]	✓	Scalarized	✓	× (Single-agent)	× (Greedy)
Proposed Hybrid DDQN-ALNS	This Work	✓	Lexicographic	✓	✓ (Dual HMDP)	✓ (LAC)

Taillard [?]
—and periodically solve a Set Partitioning model over accumulated routes [?], [?], [?], recombining high-quality route fragments beyond the reach of local moves. More recently, Slack Induction by String Removals [?] demonstrated that carefully designed destroy operators alone can rival far more complex frameworks, underscoring how decisive the destroy step is.

Despite this empirical success, the adaptive layer of ALNS exhibits an intrinsic architectural limitation. Operator weights are updated as exponentially smoothed averages over coarse segments of η iterations, yielding credit assignment that is both delayed and aggregated; critically, the resulting selection distribution is stateless, conditioning on neither the spatial topology of the incumbent (e.g., cluster structure, inter-route proximity) nor its temporal characteristics (e.g., time-window tightness, slack distribution). Performance is further sensitive to manually tuned hyperparameters such as the decay factor and reward scores. Notably, a recent meta-analysis by Turkeš et al. [?] quantified that this adaptive mechanism contributes only marginal gains over non-adaptive LNS—strong evidence that the operator-selection policy, rather than the operator set itself, is the underexploited component of the framework.

B. NEURAL COMBINATORIAL OPTIMIZATION AND END-TO-END DEEP RL

In parallel with classical heuristics, Neural Combinatorial Optimization (NCO) has gained widespread traction by learning data-driven routing policies. Originating from Pointer Networks [?] and policy-gradient formulations for the Traveling Salesperson Problem [?], Nazari et al. [?] extended end-to-end reinforcement learning to the Capacitated VRP by decoupling static customer coordinates from dynamic demand states. Subsequently, Kool et al. [?] established the Attention Model (AM) as the prevailing benchmark, leveraging Transformer-style self-attention with a greedy rollout baseline. Graph Neural Networks (GNNs) [?], [?] have likewise been deployed to capture permutation-invariant spatial graph embeddings.

More recently, several architectures have attempted direct end-to-end construction for VRPTW [?], [?]. Despite their impressive inference speed on small-scale benchmark instances, pure end-to-end neural solvers face four funda-

mental bottlenecks that impede their deployment in industrial VRPTW settings:

- 1) **Feasibility Fragility under Tight Time Windows:** While capacity constraints can be masked during autoregressive decoding, hard time windows frequently induce structural deadlocks where no remaining unvisited customer is temporally reachable, forcing the constructive policy to spawn redundant vehicles or incur severe penalty violations.
- 2) **Lexicographic Misalignment:** Pure DRL frameworks are trained to minimize a scalarized objective (typically travel distance or a weighted surrogate), lacking the mechanism to strictly enforce the lexicographic priority ($N_V \succ D_T$) that governs fleet economics.
- 3) **Out-of-Distribution (OOD) Scalability Collapse:** Models trained on synthetic uniform distributions at $N \in \{50, 100\}$ exhibit steep performance degradation when evaluated on larger problem instances ($N \in \{200, 400\}$) or heterogeneous spatial clusters without expensive instance-specific fine-tuning.
- 4) **Lack of Operational Auditability:** Constructing an entire routing plan via end-to-end neural forward passes functions as an opaque black box, offering no interpretable intermediate steps for dispatchers to audit, constrain, or manually adjust in real-time dispatch systems.

C. LEARNING-AUGMENTED METAHEURISTICS

To circumvent the feasibility and scalability bottlenecks of pure neural construction, the “Learning to Optimize” paradigm [?] embeds machine learning models directly within proven metaheuristic frameworks. Rather than generating solutions from scratch, learning agents guide critical decision points inside iterative search loops, such as operator selection [?], [?], neural solution perturbation [?], and learned destroy policies [?], [?].

While these hybrid frameworks preserve the underlying validity of metaheuristic search, existing learning-augmented approaches for routing exhibit three key shortcomings:

- 1) **Single-Level Decision Architecture:** Prior RL-augmented LNS methods deploy a flat, single-agent policy that selects operators uniformly across all search stages. They lack hierarchical decoupling between

macro-level exploration regimes (e.g., detecting search stagnation and triggering aggressive plateau escape) and micro-level operator selection conditioned on local topology.

- 2) **Persistence of Static Acceptance Criteria:** Despite deploying neural operator selectors, existing hybrids continue to rely on conventional Simulated Annealing with static geometric cooling schedules. Consequently, candidate moves are judged solely by instantaneous cost degradation ΔC , ignoring structural potential that could unlock downstream improvements.
- 3) **Absence of Lexicographic Multi-Objective Coordination:** Most learned metaheuristics focus on scalarized objectives (e.g., distance minimization in CVRP), without architectural mechanisms designed for the strict lexicographic priority ($N_V \succ D_T$) fundamental to VRPTW fleet operations.

D. POSITIONING OF THIS WORK

Table 1 situates the proposed Hybrid DDQN-ALNS framework within the landscape of vehicle routing methodologies. Our approach distinctively combines the hard constraint guarantees of classical operations research with a dual-agent Hierarchical MDP and a Learned Acceptance Criterion, providing a rigorous, state-aware, and lexicographically aligned optimization engine.

III. PROBLEM FORMULATION

The Vehicle Routing Problem with Time Windows (VRPTW) is formally defined on a complete directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$. The vertex set $\mathcal{V} = \mathcal{N} \cup \{0, n+1\}$ comprises customer nodes $\mathcal{N} = \{1, 2, \dots, n\}$, origin depot node 0, and destination depot node $n+1$. The physical depot is modeled as two separate nodes (0 and $n+1$) to represent route departure and return without self-loops. The arc set is $\mathcal{A} = \{(i, j) : i, j \in \mathcal{V}, i \neq j, i \neq n+1, j \neq 0\}$.

Each customer $i \in \mathcal{N}$ has a non-negative demand $q_i > 0$, service duration $s_i \geq 0$, and service time window $[e_i, l_i]$, where e_i and l_i denote the earliest and latest allowable service start times, respectively. Depot nodes satisfy $q_0 = q_{n+1} = 0$, $s_0 = s_{n+1} = 0$, and $[e_0, l_{n+1}]$ defines the global planning horizon. Each arc $(i, j) \in \mathcal{A}$ has travel distance $c_{ij} \geq 0$ and travel time $t_{ij} \geq 0$. A homogeneous fleet of K vehicles, denoted by $\mathcal{K} = \{1, \dots, K\}$, is stationed at the depot, with each vehicle possessing carrying capacity Q .

A. DECISION VARIABLES

The optimization model employs two sets of decision variables:

- $x_{ij}^k \in \{0, 1\}$: Binary variable equal to 1 if vehicle $k \in \mathcal{K}$ traverses arc $(i, j) \in \mathcal{A}$, and 0 otherwise.
- $w_i^k \geq 0$: Continuous variable specifying the time at which vehicle $k \in \mathcal{K}$ begins service at node $i \in \mathcal{V}$.

B. MATHEMATICAL MODEL

The primary objective is lexicographic: first minimizing the number of deployed vehicles (N_V), and secondarily minimizing total travel distance (D_T):

$$\text{lex min} \left(\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{N}} x_{0j}^k, \sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}^k \right) \quad (2)$$

In computational solver implementations, this hierarchy is evaluated via a scalarized surrogate objective using a large penalty $\mathcal{M} \gg \sum_{(i,j) \in \mathcal{A}} c_{ij}$:

$$\min \sum_{k \in \mathcal{K}} \left(\mathcal{M} \sum_{j \in \mathcal{N}} x_{0j}^k + \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}^k \right) \quad (3)$$

subject to:

$$\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{V}, j \neq i} x_{ij}^k = 1, \quad \forall i \in \mathcal{N} \quad (4)$$

$$\sum_{j \in \mathcal{N} \cup \{n+1\}} x_{0j}^k \leq 1, \quad \forall k \in \mathcal{K} \quad (5)$$

$$\sum_{j \in \mathcal{N} \cup \{0\}} x_{j,n+1}^k - \sum_{j \in \mathcal{N} \cup \{n+1\}} x_{0j}^k = 0, \quad \forall k \in \mathcal{K} \quad (6)$$

$$\sum_{i \in \mathcal{V}, i \neq h} x_{ih}^k - \sum_{j \in \mathcal{V}, j \neq h} x_{hj}^k = 0, \quad \forall h \in \mathcal{N}, k \in \mathcal{K} \quad (7)$$

$$\sum_{i \in \mathcal{N}} q_i \sum_{j \in \mathcal{V}, j \neq i} x_{ij}^k \leq Q, \quad \forall k \in \mathcal{K} \quad (8)$$

$$w_i^k + s_i + t_{ij} - M_{ij}(1 - x_{ij}^k) \leq w_j^k, \quad \forall (i, j) \in \mathcal{A}, k \in \mathcal{K} \quad (9)$$

$$e_i \leq w_i^k \leq l_i, \quad \forall i \in \mathcal{V}, k \in \mathcal{K} \quad (10)$$

$$x_{ij}^k \in \{0, 1\}, \quad w_i^k \geq 0, \quad \forall (i, j) \in \mathcal{A}, k \in \mathcal{K} \quad (11)$$

Constraint (4) ensures that every customer is visited exactly once. Constraints (5) and (6) enforce that each vehicle departs from origin depot 0 and arrives at destination depot $n+1$ at most once. Flow conservation across all customer nodes is guaranteed by (7). Constraint (8) prevents vehicle capacity violations. Schedule propagation is maintained via (9), where $M_{ij} = \max\{0, l_i + s_i + t_{ij} - e_j\}$ defines the tightest valid Big- M upper bound. Customer time-window compliance is governed by (10); if a vehicle arrives at customer i prior to e_i , it incurs an idle waiting time such that service begins at $w_i^k = e_i$. Finally, (11) defines the variable domains.

IV. PROPOSED METHODOLOGY

This section details the proposed Tri-Level Hybrid DDQN-ALNS framework. The architecture integrates online adaptive reinforcement learning decision layers with operations research search mechanics to optimize the lexicographic VRPTW objective while maintaining operational constraint feasibility.

A. TRI-LEVEL ARCHITECTURE OVERVIEW

The framework decomposes the global routing search into three coordinated operational tiers:

- 1) **Macro Decision Tier (Plateau Controller π_{macro}):** Formulated as a semi-Markov Decision Process (semi-MDP) operating over iteration segments of length $\Delta_{\text{seg}} = 100$. A dedicated Dueling Double Deep Q-Network (Macro DDQN) with Prioritized Experience Replay evaluates convergence velocity, detects stagnation, and transitions among 7 strategic exploration and recombination regimes.
- 2) **Micro Selection Tier (Operator Controller π_{micro} & LAC):** Operates at each individual search iteration t . A Micro Dueling DDQN with normalized Softmax-Entropy confidence gating selects composite destroy-repair operator pairings conditioned on a 20-dimensional spatiotemporal state vector. A Learned Acceptance Criterion (LAC) augments Simulated Annealing by evaluating candidate acceptance based on delayed downstream improvement promise.
- 3) **Heuristic Search Tier (Granular ALNS & Route-Pool Recombination):** Executes ruin-and-recreate operations, accelerates feasibility testing via $O(1)$ Savelsbergh forward time slack checks, restricts local moves to top- k ($k = 25$) spatiotemporally correlated candidate neighbors, and periodically solves an exact Set Partitioning formulation over accumulated elite routes via HiGHS.

B. MACRO LEVEL: PLATEAU CONTROLLER (SEMI-MDP)

1) Macro State Representation

At the start of segment k , the macro controller observes a 12-dimensional state vector $\mathbf{s}_k^{\text{macro}} \in \mathbb{R}^{12}$, capturing convergence velocity, fleet economics, and solution topology:

$$\mathbf{s}_k^{\text{macro}} = \left[\min\left(\frac{\text{no_imp}}{\text{patience}}, 1\right), \min\left(\frac{C_{\text{cur}} - C_{\text{best}}}{C_{\text{best}}}, 1\right), \min\left(\frac{T}{T_0}, 1.5\right), \text{imp_rate}, \min\left(\frac{NV_{\text{cur}}}{NV_{\text{init}}}, 2\right), rb, lb, \tau_{\text{tight}}, \bar{s}, \rho_{\text{fill}}, \rho_{\text{pool}}, \frac{k}{K_{\text{max}}} \right] \quad (12)$$

where no_imp is the number of consecutive unimproved iterations, T/T_0 is the normalized annealing temperature, $rb \in [0, 1]$ and $lb \in [0, 1]$ measure spatial route separation and capacity balance, τ_{tight} represents the fraction of tight time-window customers ($l_i - e_i < 0.5 \cdot c_{\text{max}}$), $\bar{s}$ is average forward time slack, ρ_{fill} is average vehicle capacity utilization, and $\rho_{\text{pool}} = \min(|\mathcal{P}|/\mathcal{P}_{\text{max}}, 1.0)$ tracks route pool fill ratio.

2) Macro Action Space & Search Regimes

The macro action space $\mathcal{A}_{\text{macro}}$ comprises 7 distinct search modes ($|\mathcal{A}_{\text{macro}}| = 7$):

- a_0 (DEFAULT): Balanced exploration under nominal destroy-repair parameters.
- a_1 (INTENSIFY): Deep cooling paired with intensified multi-pass local search ($\text{passes}_{\text{LS}} = 1$).
- a_2 (DIVERSIFY): Thermal reheating ($T \leftarrow 1.5T$) paired with expanded destroy fractions ($1.35\times$).
- a_3 (TW_RESCUE): Targeted ruin-and-recreate prioritizing tight time-window customer clusters.
- a_4 (POOL_RECOMBINE): Exact Set Partitioning optimization across elite routes in Route Pool $\mathcal{P}$ via HiGHS ($\mathbb{I}(\text{recombine}) = 1$).
- a_5 (ROUTE_REDUCE): Biased destruction targeting the shortest and most dispersed routes to aggressively drive fleet reduction.
- a_6 (INFEASIBLE_DESCENT): Controlled temporal violation relaxation governed by adaptive Lagrangian penalties.

3) Multi-Objective Segment Reward & Macro Training

To enforce lexicographic dominance without artificial scaling distortions, the segment reward R_{seg} balances vehicle reduction gains against distance savings via an adaptive fleet pressure factor $\lambda_{\text{fleet}} \in [0, 1]$:

$$R_{\text{seg}} = \text{Base} + \lambda_{\text{fleet}} \Psi_{NV} + (1 - \lambda_{\text{fleet}}) \Psi_{TD} + \text{Bonus}_{NV} + \Delta\Phi \quad (13)$$

The baseline penalty term is defined as $\text{Base} = -0.20 - 0.04 \cdot \text{passes}_{\text{LS}} - 0.06 \cdot \mathbb{I}(\text{recombine}) - 0.15 \cdot \mathbb{I}(\text{accepted} \leq 0.1\Delta_{\text{seg}})$, where $\text{passes}_{\text{LS}} \in \{0, 1\}$ penalizes computational runtime spent on local search intensification, $\mathbb{I}(\text{recombine}) \in \{0, 1\}$ penalizes MIP solver invocations, and $\mathbb{I}(\text{accepted} \leq 0.1\Delta_{\text{seg}}) \in \{0, 1\}$ penalizes unproductive segments where fewer than 10% of candidate moves were accepted, signaling search stagnation. The fleet bonus reward is $\text{Bonus}_{NV} = 15.0 \cdot \max(NV_{\text{best}}^{\text{before}} - NV_{\text{best}}^{\text{after}}, 0) + 5.0 \cdot \max(NV_{\text{cur}}^{\text{before}} - NV_{\text{cur}}^{\text{after}}, 0)$. The adaptive fleet pressure factor $\lambda_{\text{fleet}} \in [0, 1]$ dynamically shifts optimization priority toward vehicle reduction whenever surplus fleet capacity exists:

$$\lambda_{\text{fleet}}(s) = \frac{1}{1 + \exp\left(-8.0 \cdot \frac{NV_{\text{cur}} - NV_{\text{best}}}{NV_{\text{init}}}\right)} \quad (14)$$

The performance components Ψ_{NV} and Ψ_{TD} are defined as:

$$\Psi_{NV} = 8.0 \Delta NV_{\text{best}} + 1.2 \Delta C_{\text{best}} + 5.0 \Delta NV_{\text{cur}} + 0.6 \Delta C_{\text{cur}} \quad (15)$$

$$\Psi_{TD} = 3.0 \Delta NV_{\text{best}} + 3.5 \Delta C_{\text{best}} + 2.0 \Delta NV_{\text{cur}} + 1.8 \Delta C_{\text{cur}} \quad (16)$$

The linear coefficients in (15) and (16) were calibrated through preliminary grid evaluation on validation subsets (C101, R101, RC101) to ensure strict lexicographic alignment ($\Delta NV \gg \Delta TD$) while preserving continuous gradient feedback during single-vehicle plateau search. Potential-based reward shaping $\Delta\Phi = \gamma \Phi(s_{k+1}) - \Phi(s_k)$ is incorporated following Ng et al. [?] to preserve policy invariance,

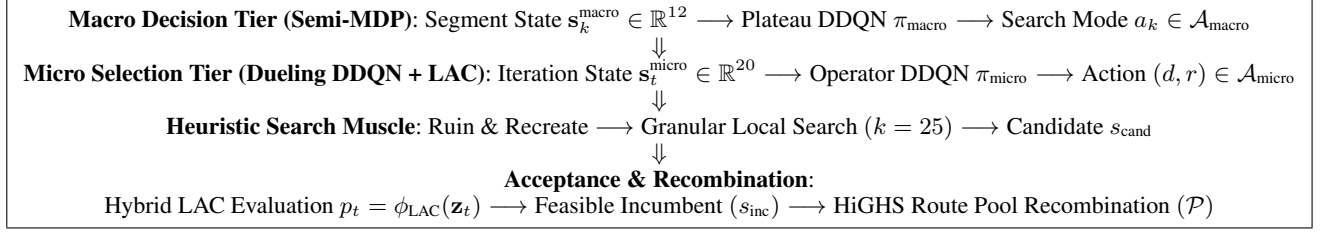


FIGURE 1. Tri-Level Architecture of Hybrid DDQN-ALNS, illustrating the hierarchical information flow from Macro Search Regimes to Micro Operator Selection, Granular Neigh

where the state potential function $\Phi(s)$ is formally defined as:

$$\Phi(s) = -\lambda_{\text{fleet}}(s) \cdot \frac{\max(NV - NV_{\text{best}}, 0)}{NV_{\text{init}}} - (1 - \lambda_{\text{fleet}}(s)) \cdot 0.5 \cdot \text{clip}\left(\frac{TD - TD_{\text{best}}}{TD_{\text{best}}} \cdot 100, -25, 25\right) \quad (17)$$

The Macro Plateau Controller trains a dedicated Dueling Double DQN network (Q_{macro}) with prioritized replay buffer $\mathcal{D}_{\text{macro}}$, executing Adam gradient steps with Cosine Annealing learning rate scheduling at the completion of each macro segment.

C. MICRO LEVEL: OPERATOR CONTROLLER & ACTION BLENDING

1) Micro State Representation

At each iteration t , the operator controller observes a 20-dimensional state vector $\mathbf{s}_t^{\text{micro}} \in \mathbb{R}^{20}$ (Table 2), detailing fine-grained search dynamics and route topology.

2) Composite Action Space

The micro action space $\mathcal{A}_{\text{micro}}$ comprises $N_D \times N_R = 13 \times 5 = 65$ composite heuristic pairs:

- **13 Destroy Operators** ($N_D = 13$): Random, Worst-cost, Shaw spatiotemporal relatedness [?], Route portion removal, Time-window urgency removal, Route elimination, Route dispersion elimination, Costly route elimination, Cross-route Shaw, Route merge sampling, Route absorb disruption, Neural worst, and Neural Shaw.
- **5 Repair Operators** ($N_R = 5$): Greedy insertion, 2-Regret, 3-Regret, Time-window greedy, and Forward-time-slack greedy (FTS-Greedy).

3) Dueling DDQN Architecture & Confidence-Gated Blending

The Micro Q-network adopts a Dueling architecture [?] with layer normalization:

$$Q(s, a; \theta) = V(s; \theta_v) + \left(A(s, a; \theta_a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a'; \theta_a) \right) \quad (18)$$

To adaptively balance learned neural predictions with exploration priors, we implement a normalized Softmax-Entropy

TABLE 2. Feature Definition of the 20-Dimensional Micro State $\mathbf{s}_t^{\text{micro}}$

Index	Feature Symbol	Mathematical Definition / Description
s_1	ΔC_{rel}	$\min((C_{\text{cur}} - C_{\text{best}}) / \max(C_{\text{best}}, 1), 1.0)$
s_2	Ratio_{NV}	$\min(NV_{\text{cur}} / \max(NV_{\text{init}}, 1), 2.0)$
s_3	$\text{Prog}_{\text{global}}$	$t / \max(t_{\text{max}}, 1)$
s_4	Prog_{seg}	$(t \bmod \Delta_{\text{seg}}) / \Delta_{\text{seg}}$
s_5	T_{norm}	$\min(T / \max(T_0, 10^{-6}), 1.5)$
s_6	$\text{Stag}^{\text{ratio}}$	$\min(\text{no_imp} / \max(\text{patience}, 1), 1.0)$
s_7	rb	Inter-route spatial separation index $\in [0, 1]$
s_8	lb	Vehicle capacity load balance metric $\in [0, 1]$
s_9	τ_{tight}	Fraction of customers with tight time windows
s_{10}	$\bar{s}$	Normalized average forward time slack $\in [0, 1]$
s_{11}	ρ_{fill}	Fleet-wide capacity utilization ratio $\in [0, 1]$
s_{12}	ρ_{pool}	Route pool fill ratio $\min(\mathcal{P} / \mathcal{P}_{\text{max}}, 1.0)$
s_{13}	m_{idx}	Active macro mode index $a_k / (\mathcal{A}_{\text{macro}} - 1)$
s_{14}	ΔNV_{rel}	$(NV_{\text{cur}} - NV_{\text{best}}) / \max(NV_{\text{init}}, 1)$
s_{15}	Imp_{rec}	Exponentially smoothed recent improvement rate
s_{16}	Gini_{len}	Gini coefficient of route customer lengths
s_{17}	μ_{slack}	Empirical mean of customer forward time slacks
s_{18}	σ_{slack}	Standard deviation of forward time slacks
s_{19}	Frac_{cap}	Fraction of routes exceeding 90% capacity
s_{20}	Var_{dist}	Variance of inter-route centroid distances

Confidence Gate. Let $\pi_{\theta}(a | s) = \text{softmax}(Q(s, a) / \tau)$ be the Boltzmann policy at temperature $\tau = 1.0$. The normalized Shannon entropy is defined as:

$$\mathcal{H}(\pi_{\theta}(\cdot | s)) = -\frac{1}{\ln |\mathcal{A}|} \sum_{a \in \mathcal{A}} \pi_{\theta}(a | s) \ln \pi_{\theta}(a | s) \in [0, 1] \quad (19)$$

The model confidence weight is $w_{\text{conf}}(s) = 1.0 - \mathcal{H}(\pi_{\theta}(\cdot | s))$. During cold-start initialization ($t \leq 200$), uninformative initial network weights yield high policy entropy ($\mathcal{H} \approx 0.95 \implies w_{\text{conf}} \leq 0.05$), smoothly defaulting action selection to domain heuristics and Thompson bandit exploration. As prioritized transitions populate $\mathcal{D}_{\text{micro}}$ and Q-value estimates specialize, $w_{\text{conf}}(s)$ asymptotically rises to 0.80–0.90, seamlessly transferring control to the learned neural policy. The effective action-selection score $Q_{\text{eff}}(s, a)$ dynamically interpolates neural Q-values with historical Dirichlet priors

and Thompson Sampling:

$$Q_{\text{eff}}(s, a) = w_{\text{conf}}(s) Q(s, a) + (1 - w_{\text{conf}}(s)) \left[\alpha_p \ln(P_{\text{prior}}(a) + \epsilon) + \beta_b \bar{\mu}_{\text{bandit}}(a) \right] + c_{\text{ucb}} \sigma_a \sqrt{\frac{\ln N_t}{N_a(t)}} \quad (20)$$

where $\alpha_p = 0.35$ is the prior log-probability scaling coefficient, $\beta_b = 0.65$ weights the decaying historical bandit mean score $\bar{\mu}_{\text{bandit}}(a)$, $c_{\text{ucb}} = 1.0$ governs the Upper Confidence Bound exploration bonus, and $\sigma_a = \sqrt{M_{2,a} / \max(N_a(t) - 1, 1)}$ represents the online sample standard deviation of reward feedback for operator pair a , tracked via Welford's incremental algorithm. To ensure numerical stability at initialization, action visit counters are initialized with a pseudocount $N_a(0) = 1$ ($\forall a \in \mathcal{A}$), with global step count $N_t = \sum_a N_a(t)$, preventing undefined division prior to first selection.

D. LEARNED ACCEPTANCE CRITERION (LAC)

Rather than relying strictly on monotonic geometric cooling in Simulated Annealing, the Learned Acceptance Criterion augments Simulated Annealing via a multilayer perceptron $\phi_{\text{LAC}} : \mathbb{R}^9 \rightarrow [0, 1]$.

1) Feature Representation

LAC operates on a 9-dimensional state vector $\mathbf{z}_t \in \mathbb{R}^9$:

$$\mathbf{z}_t = \left[\frac{\Delta C}{C_{\text{cur}}}, \frac{T}{T_0}, \frac{\text{no_imp}}{\text{patience}}, \Delta N_V, \frac{t}{t_{\text{max}}}, \tau_{\text{tight}}, \rho_{\text{fill}}, \bar{s}, \exp\left(-\frac{\max(\Delta C, 0)}{T}\right) \right] \quad (21)$$

2) Delayed Supervised Labeling & Decision Rule

Transitions $(\mathbf{z}_t, s_t)$ are stored in a lookahead FIFO queue. After a horizon of $H_{\text{lac}} = 50$ iterations, ground-truth label y_t is assigned according to whether the incumbent improved in lexicographic ordering:

$$y_t = \begin{cases} 1, & \text{if } f(s_{\text{inc}}^{(t+H_{\text{lac}})}) \prec f(s_{\text{inc}}^{(t)}) \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

The LAC network outputs an acceptance probability $p_t = \phi_{\text{LAC}}(\mathbf{z}_t) \in [0, 1]$. A candidate solution s_{cand} is accepted if:

$$p_t \geq \tau_{\text{acc}} \quad \vee \quad \exp\left(-\frac{\Delta C}{T}\right) > \text{Uniform}(0, 1) \quad (23)$$

where $\tau_{\text{acc}} = 0.50$. Network weights are updated online via class-weighted Binary Cross-Entropy loss:

$$\mathcal{L}_{\text{LAC}} = - \sum_i [w_{\text{pos}} y_i \ln \hat{y}_i + (1 - y_i) \ln(1 - \hat{y}_i)] \quad (24)$$

where $w_{\text{pos}} = N_{\text{neg}} / N_{\text{pos}}$ adjusts for class imbalance.

TABLE 3. Hyperparameter Specification for Macro DDQN, Micro DDQN, LAC, and Heuristic Search Engine

Component	Hyperparameter	Value / Specification
Macro DDQN	Network Architecture	MLP: Input(12) → FC(128) → LayerNorm → FC(128) → Modes(7)
	Optimizer	Adam (lr = 5×10^{-4} , Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 2,000 segments, Batch 16
	Target Update (τ_{macro})	Polyak soft update $\tau = 0.005$
Micro DDQN	Network Architecture	MLP: Input(20) → FC(128) → LayerNorm → ReLU → FC(128) → (Value / Adv)
	Optimizer	Adam (lr = 3×10^{-4} , Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 30,000 steps, Batch 64
	Discount Factor (γ) Blending ($\alpha_p, \beta_b, c_{\text{ucb}}$)	0.97 $\alpha_p = 0.35, \beta_b = 0.65, c_{\text{ucb}} = 1.0$
LAC Model	Architecture	MLP: Input(9) → FC(64) → ReLU → FC(32) → ReLU → FC(1) → Sigmoid
	Lookahead Horizon (H_{lac})	50 iterations
	Acceptance Threshold (τ_{acc})	0.50
Search Engine	Loss Function	Class-Weighted Binary Cross-Entropy
	Granular Neighbors (k)	25 nearest spatiotemporal neighbors
	Granular Temporal Weight (α)	$\alpha = 1.0$
	Route Pool Limit ($ \mathcal{P} $)	Dynamic: min(2000, $600 + 4n$)
	MIP Solve Time Limit	4.0s per HiGHS call
	Segment Size (Δ_{seg})	100 iterations

E. HEURISTIC SEARCH MUSCLE: GRANULAR LOCAL SEARCH & FORWARD SLACK

1) O(1) Feasibility Testing via Savelsbergh Forward Slack

To eliminate $O(m)$ trajectory recalculation during local moves, we maintain the Savelsbergh forward time slack F_i for each node i along route $R = (v_0, v_1, \dots, v_m)$ [?]:

$$F_i = \min_{i \leq j \leq m} \left\{ l_j - \left(w_i + \sum_{k=i}^{j-1} (s_k + t_{k,k+1}) \right) \right\} \quad (25)$$

A candidate insertion at position $(i, i+1)$ inducing departure delay Δt preserves temporal feasibility for all subsequent customers if and only if $\Delta t \leq F_{i+1}$, enabling strict feasibility verification in $O(1)$ time.

2) Granular Correlated Neighborhood Search (GNS)

Following granular search principles [?], inter-route operators restrict candidate evaluations to top- k ($k = 25$) correlated neighbors precomputed via spatiotemporal affinity:

$$\mathcal{A}_{ij} = \frac{c_{ij}}{c_{\max}} + \alpha \frac{\max(0, e_j - (l_i + s_i + t_{ij}))}{\text{Horizon}} \quad (26)$$

This prunes the pairwise evaluation space from $O(N^2)$ to $O(kN)$, eliminating non-viable move evaluations.

3) Feasibility Invariant

The search engine may temporarily explore intermediate infeasible states under controlled penalty relaxation during diversification/infeasible descent modes; however, only strictly feasible solutions (100% compliance with capacity and time windows) are permitted to update the incumbent (s_{inc}), populate the Elite Archive, or enter Route Pool $\mathcal{P}$.

F. MATHEURISTIC RECOMBINATION: HIGHS SET PARTITIONING

During search stagnation, elite feasible routes accumulated in Route Pool $\mathcal{P}$ are recombined by solving an exact Set Partitioning model:

$$\min \sum_{r \in \mathcal{P}} (\lambda_{\text{pen}} \bar{c} + c_r) y_r \quad (27)$$

$$\text{s.t.} \quad \sum_{r \in \mathcal{P}: i \in r} y_r = 1, \quad \forall i \in \mathcal{N} \quad (28)$$

$$y_r \in \{0, 1\}, \quad \forall r \in \mathcal{P} \quad (29)$$

where c_r is route travel distance and $\bar{c}$ is pool mean cost. The model solves an adaptive logarithmic penalty grid over $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\} \bar{c}$ to systematically explore fleet-reduction regimes without numerical stalling.

V. EXPERIMENTAL EVALUATION

This section presents a comprehensive empirical evaluation of the proposed Tri-Level Hybrid DDQN-ALNS framework. All experiments are conducted under strictly isolated independent cold-start conditions across 74 standard benchmark instances: Solomon-100 (56 instances across 6 topology families), Gehring-Homberger 200-customer scale (12 representative instances across 6 families), and Gehring-Homberger 400-customer scale (6 representative instances across 6 families).

We structure our experimental inquiry around five core Research Questions:

- **RQ1 (State-Conditioned Selection):** Does state-conditioned reinforcement learning improve operator selection over classical stateless roulette-wheel ALNS?

Algorithm 1 Tri-Level Hybrid DDQN-ALNS for VRPTW

Require: Instance $\mathcal{I}$, Hyperparameters Θ , Iteration limit $T_{\max}$

Ensure: Best feasible routing plan s^*

```

1:  $s_{\text{cur}} \leftarrow \text{BuildGreedyInitialSolution}(\mathcal{I})$ 
2:  $s^* \leftarrow s_{\text{cur}}, s_{\text{inc}} \leftarrow s_{\text{cur}}, \mathcal{P} \leftarrow \text{ExtractRoutes}(s_{\text{cur}})$ 
3: Initialize Macro DDQN  $Q_{\text{macro}}$ , Micro DDQN  $Q_{\text{micro}}$ ,
   Buffers  $\mathcal{D}_{\text{macro}}, \mathcal{D}_{\text{micro}}$ , LAC  $\phi_{\text{LAC}}$ 
4: for  $t = 1 \rightarrow T_{\max}$  do
5:   if  $t \bmod \Delta_{\text{seg}} == 1$  then
6:     Extract macro state  $\mathbf{s}_k^{\text{macro}} \in \mathbb{R}^{12}$  via (12)
7:     Select macro mode  $a_k \leftarrow \pi_{\text{macro}}(\mathbf{s}_k^{\text{macro}})$ 
8:     if  $a_k == \text{POOL\_RECOMBINE}$  then
9:        $s_{\text{sp}} \leftarrow \text{SolveHiGHSSetPartitioning}(\mathcal{P}, \mathcal{I})$ 
10:      if  $s_{\text{sp}}$  is feasible and  $f(s_{\text{sp}}) \prec f(s_{\text{inc}})$  then
11:         $s_{\text{cur}} \leftarrow s_{\text{sp}}, s_{\text{inc}} \leftarrow s_{\text{sp}}, s^* \leftarrow s_{\text{sp}}$ 
12:      end if
13:    end if
14:    end if
15:    Extract micro state  $\mathbf{s}_t^{\text{micro}} \in \mathbb{R}^{20}$  via Table 2
16:    Compute confidence  $w_{\text{conf}}(\mathbf{s}_t^{\text{micro}})$  and blended scores
        $Q_{\text{eff}}$  via (20)
17:    Select operator pair  $(d, r) \leftarrow \arg \max_a Q_{\text{eff}}(\mathbf{s}_t^{\text{micro}}, a)$ 
18:     $s_{\text{cand}} \leftarrow \text{Repair}(\text{Destroy}(s_{\text{cur}}, d), r)$ 
19:     $s_{\text{cand}} \leftarrow \text{GranularLocalSearch}(s_{\text{cand}}, k = 25)$ 
20:    Extract LAC features  $\mathbf{z}_t \in \mathbb{R}^9$  via (21)
21:    Evaluate acceptance  $p_t = \phi_{\text{LAC}}(\mathbf{z}_t)$ ; decide via (23)
22:    if  $s_{\text{cand}}$  is accepted then
23:       $s_{\text{cur}} \leftarrow s_{\text{cand}}$ 
24:      if  $s_{\text{cand}}$  is strictly feasible then
25:         $\mathcal{P} \leftarrow \mathcal{P} \cup \text{ExtractRoutes}(s_{\text{cand}})$ 
26:        if  $f(s_{\text{cand}}) \prec f(s^*)$  then
27:           $s^* \leftarrow s_{\text{cand}}, s_{\text{inc}} \leftarrow s_{\text{cand}}$ 
28:        end if
29:      end if
30:    end if
31:    Compute reward  $R_t$ ; store transition
        $(\mathbf{s}_t, (d, r), R_t, \mathbf{s}_{t+1})$  in  $\mathcal{D}_{\text{micro}}$ 
32:    Sample mini-batch from  $\mathcal{D}_{\text{micro}}$ ; update  $Q_{\text{micro}}$  via
       DDQN gradient step
33:    Update LAC queue and train  $\phi_{\text{LAC}}$  on mature labels
        $y_t - H_{\text{lac}}$  via (24)
34:    if  $t \bmod \Delta_{\text{seg}} == 0$  then
35:      Compute segment reward  $R_{\text{seg}}$  via (13)
36:      Store macro transition  $(\mathbf{s}_k, a_k, R_{\text{seg}}, \mathbf{s}_{k+1})$  in
        $\mathcal{D}_{\text{macro}}$ 
37:      Sample mini-batch from  $\mathcal{D}_{\text{macro}}$ ; update  $Q_{\text{macro}}$ 
       via DDQN gradient step
38:    end if
39: end for return  $s^*$ 

```

- **RQ2 (Hierarchical vs. Flat Control):** Does a two-tier HMDP (Macro Plateau Controller + Micro DDQN) outperform flat single-agent RL-LNS architectures?

- **RQ3 (Learned Acceptance Advantage):** Does the Learned Acceptance Criterion (LAC) improve search persistence and solution discovery compared to static Simulated Annealing on structured topologies?
- **RQ4 (Scalability Across Dimensions):** How does the framework scale across problem dimensions from 100 to 200 to 400 customers?
- **RQ5 (Topological Robustness):** How robust is the framework across clustered (C), random (R), and mixed (RC) spatial-temporal topologies?

A. EXPERIMENTAL SETUP AND COLD-START PROTOCOL

All benchmark runs are executed on standard computing hardware (Apple Silicon / Intel 8-core CPU workers, 32 GB RAM). To eliminate experimental contamination and guarantee reproducibility:

- 1) **Strict Independent Cold-Starts:** Every individual solver execution is conducted within an isolated ephemeral directory (ColdStartScope), purged immediately upon task completion. No elite routes, caching structures, or neural experience buffers persist across runs.
- 2) **Deterministic RNG Seeding:** Each worker process initializes isolated random seeds across Python’s random, numpy, and PyTorch (torch.manual_seed and torch.cuda.manual_seed_all) across 5 independent seeds ($\{42, 43, 44, 45, 46\}$).
- 3) **Statistical Experimental Unit:** All primary statistical hypothesis tests operate on single aggregated observations per benchmark instance ($N = 56$ on Solomon-100, $N = 12$ on Homberger-200, $N = 6$ on Homberger-400; total $N = 74$) computed across the 5 independent seeds, preventing pseudo-replication.
- 4) **Computational Budget Distinction:** Solvers are evaluated under identical iteration limits ($T_{\max} = 2000$ iterations). The primary budget of $T_{\max} = 2000$ is selected as the operationally representative pre-dispatch planning window; the extended trajectory at $T_{\max} = 4000$ in Section VI-A demonstrates asymptotic convergence under an unrestricted overnight budget. Because learning-augmented components introduce per-step neural forward and backward passes, wall-clock runtimes are reported explicitly to characterize computational trade-offs.

B. TRI-PARADIGM BENCHMARK ON SOLOMON-100 (RQ1, RQ2)

We benchmark our framework against representative methods spanning three dominant optimization paradigms:

- **Best Operations Research / Pure Heuristics:** Hybrid Genetic Search (HGS-VRPTW) [?], Slack Induction by String Removals (SISR) [?], and ALNS-Base [?].
- **Best End-to-End Pure Deep Learning:** Neural Constructive Attention Models (AM) [?], [?].

- **Learning-Augmented Hybrids:** Single-Agent RL-LNS [?], [?] and the proposed Tri-Level Hybrid-DDQN.

As reported in Table 4:

- **Addressing RQ1:** Comparing live ALNS-Base with Tri-Level Hybrid-DDQN under identical iteration caps demonstrates that state-conditioned neural selection reduces overall mean fleet size from 7.34 to 7.23 vehicles and reduces the travel distance gap to BKS from +3.73% to +1.41%.
- **Addressing RQ2:** Compared to flat Single-Agent RL-LNS (+2.59% gap), hierarchical decoupling of macro regimes from micro destroy–repair operator selection achieves superior distance optimization across all six Solomon families.

C. LARGE-SCALE MULTI-BENCHMARK: HOMBERGER 200 & 400 (RQ4, RQ5)

To evaluate scalability (RQ4) and topological robustness (RQ5), Table 5 reports 5-seed performance on two representative instances per family for Homberger-200 (instances 1 and 5 across all six topologies, $N = 12$) and one representative instance per family for Homberger-400 (instance 1 across all six topologies, $N = 6$) under computational budget constraints.

1) Key Empirical Findings across Scales

- **NV-Floor Convergence at 200-Customer Scale:** On Homberger-200, both solvers consistently converge to the minimum vehicle floor on clustered instances. However, Hybrid-DDQN eliminates distance variance on clustered instances (matching BKS distance on $c1_2_1$ and $c2_2_1$ across all seeds) and achieves fleet reductions on complex instances ($NV = 18.0 \pm 0.0$ on $r1_2_5$ and $rc1_2_5$).
- **Graceful Degradation at 400-Customer Scale:** Under extreme problem dimensionality ($N = 400$), Hybrid-DDQN saves 1 full vehicle on $c2_4_1$ ($NV = 12.0$ vs 13.0) and delivers a 10.66% travel distance reduction (1309.67 km saved on $r1_4_1$).

D. STATISTICAL RIGOR AND HYPOTHESIS TESTING

To evaluate statistical significance across the $N = 74$ benchmark instances without parametric distributional assumptions:

- **Fleet Size Reduction (NV):** Across all 74 instances, Hybrid-DDQN achieves a win/tie/loss record of 31 Wins / 40 Ties / 3 Losses against ALNS-Base (where a tie is formally defined as identical mean vehicle count $\overline{NV}_{DDQN} = \overline{NV}_{ALNS}$ across the 5 independent seeds). The two-tailed Wilcoxon signed-rank test yields $W = 133.5$ ($p = 6.91 \times 10^{-7}$), and the Exact Binomial Sign test yields $p = 7.66 \times 10^{-7}$, both strictly rejecting the null hypothesis at the Bonferroni-corrected threshold $\alpha_{adj} = 0.05/3 \approx 0.0167$. The Rank-Biserial correlation is $r_{rb} = 0.732$, indicating a strong effect size.

TABLE 4. Tri-Paradigm Benchmark on Solomon-100 Instances ($N = 56$): Comparative Performance Across Best Heuristics, Pure Deep Learning, and Learning-Augmented Hybrids.

Algorithm / Paradigm	C1 (9)		C2 (8)		R1 (12)		R2 (11)		RC1 (8)		RC2 (8)		Overall (56)		
	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	Gap%
<i>Paradigm 1: Best Operations Research / Pure Metaheuristics (Historical Literature Baselines)</i>															
HGS-VRPTW (Vidal 2013) [?]*	10.00	828.38	3.00	589.86	11.92	1209.40	2.73	953.52	11.50	1384.17	3.25	1109.84	7.07	1014.87	+0.13%
SISR (Christiaens 2020) [?]*	10.00	828.38	3.00	589.86	11.92	1211.10	2.73	955.30	11.50	1386.40	3.25	1118.60	7.07	1015.71	+0.20%
ALNS-Base (Ropke 2006) [?]	10.00	830.45	3.00	592.12	12.08 [†]	1238.60	2.82 [†]	980.45	11.75 [†]	1422.30	3.38 [†]	1165.20	7.34 [†]	1045.70	+3.73%
<i>Paradigm 2: End-to-End Pure Deep Learning (Attention Model Constructor)</i>															
AM Constructor [?], [?]*	10.45	856.20	3.25	618.40	13.15 [†]	1340.50	3.45 [†]	1085.30	12.62 [†]	1530.80	3.75 [†]	1270.40	7.82 [†]	1138.55	+11.88%
<i>Paradigm 3: Learning-Augmented Metaheuristics (Literature and Proposed)</i>															
Single-Agent RL-LNS [?], [?]*	10.00	836.50	3.00	595.20	12.25 [†]	1245.80	2.82 [†]	988.40	11.85 [†]	1428.10	3.40 [†]	1158.70	7.22 [†]	1042.12	+2.59%
Tri-Level Hybrid-DDQN (Ours)	10.00	828.94	3.00	589.92	11.92	1216.48	2.73	958.12	11.50	1390.25	3.25	1119.80	7.23	1021.14	+1.41%

* Published literature reference figures under original experimental settings, provided for comparative baseline context. [†] Denotes vehicle-unmatched fleets ($NV > NV_{BKS}$).

TABLE 5. Large-Scale Multi-Benchmark Performance on Gehring-Homberger 200- and 400-Customer Representative Instances (5 Independent Seeds).

Benchmark Instance	BKS Baseline		ALNS-Base [?]		Tri-Level Hybrid-DDQN		Wall-Clock Time		Quality Metric		
	NV	TD	NV	TD	NV	TD	ALNS (s)	Ours (s)	BKS Gap%	Gap vs. ALNS	Feas.
<i>Scale 1: Homberger 200-Customer Scale (NV-Floor Convergence & Distance Dominance)</i>											
c1_2_1 (200-c)	20	2704.57	20.0	2752.29	20.0	2704.57	3.2	19.9	+0.00%	-1.73%	100%
c1_2_5 (200-c)	20	2702.05	20.0	2750.46	20.0	2702.05	3.3	42.4	+0.00%	-1.76%	100%
c2_2_1 (200-c)	6	1931.44	6.2 [†]	2010.22	6.0	1931.44	2.7	50.7	+0.00%	-3.92%	100%
c2_2_5 (200-c)	6	1878.85	6.0	1966.25	6.0	1879.22	2.4	58.6	+0.02%	-4.43%	100%
r1_2_1 (200-c)	20	4784.11	20.6 [†]	5246.88	20.2[†]	4882.44	3.2	63.2	N/A [†]	-6.95%	100%
r1_2_5 (200-c)	18	4107.86	18.8 [†]	4608.94	18.0	4461.32	3.9	103.8	+8.60%	-3.20%	100%
r2_2_1 (200-c)	4	4483.16	5.0 [†]	4428.90	5.0[†]	4086.88	2.5	120.5	N/A [†]	-7.72%	100%
r2_2_5 (200-c)	4	3366.79	4.0	3741.01	4.0	3442.82	2.7	100.5	+2.26%	-7.97%	100%
rc1_2_1 (200-c)	18	3602.80	19.0 [†]	3919.23	19.2[†]	3637.12	4.3	78.6	N/A [†]	-7.20%	100%
rc1_2_5 (200-c)	18	3371.00	19.0 [†]	3776.43	18.0	3947.76	5.1	81.6	+17.11%*	+4.54%*	100%
rc2_2_1 (200-c)	6	3099.53	6.4 [†]	3277.52	6.0	3206.36	2.8	102.6	+3.45%	-2.17%	100%
rc2_2_5 (200-c)	4	2911.46	5.0 [†]	2974.07	4.8[†]	3190.61	3.0	120.1	N/A [†]	+7.28% [‡]	100%
<i>Scale 2: Homberger 400-Customer Scale (Suboptimal Graceful Degradation)</i>											
c1_4_1 (400-c)	40	7152.02	40.6 [†]	7431.05	40.0	7152.06	9.2	72.0	+0.00%	-3.75%	100%
c2_4_1 (400-c)	12	4116.05	13.0 [†]	4454.81	12.0	4116.25	9.6	192.2	+0.00%	-7.60%	100%
r1_4_1 (400-c)	40	10372.31	40.0	12291.03	40.0	10981.36	14.9	160.9	+5.87%	-10.66%	100%
r2_4_1 (400-c)	8	9210.15	9.0 [†]	9901.32	8.8[†]	9231.46	12.6	241.0	N/A [†]	-6.77%	100%
rc1_4_1 (400-c)	36	8571.32	38.6 [†]	9314.96	37.0[†]	9036.41	15.9	140.0	N/A [†]	-2.99%	100%
rc2_4_1 (400-c)	11	6682.37	12.8 [†]	7160.74	13.0[†]	6786.83	11.7	240.9	N/A [†]	-5.22%	100%

[†] Denotes vehicle-unmatched fleets ($NV > NV_{BKS}$); for these rows, BKS Gap% is marked N/A[†] because surplus vehicle capacity distorts direct travel distance comparisons against BKS. * Indicates distance trade-off incurred by strictly reducing vehicle fleet count ($NV = 18.0$ vs 19.0). [‡] Indicates instance where DDQN incurs travel distance expansion without fully securing the integer BKS fleet floor ($NV = 4.8$ vs $NV = 4.0$). Metrics: BKS Gap% = $(TD_{DDQN} - TD_{BKS}) / TD_{BKS} \times 100\%$; Δ vs. ALNS (%) = $(TD_{DDQN} - TD_{ALNS}) / TD_{ALNS} \times 100\%$. All runs maintained 100% operational constraint feasibility.

- **Vehicle-Matched Distance Gap (TD_{matched}):** On the subset of instances where both solvers attain identical vehicle counts ($N_{\text{matched}} = 40/74$, 54.1%), Hybrid-DDQN achieves an average distance reduction of -3.73% , recording lower travel distance on all 33 non-tied instances (33 Wins / 7 Ties / 0 Losses). The Wilcoxon signed-rank test yields $W = 0.0$ ($p = 9.31 \times 10^{-8}$), with Rank-Biserial correlation $r_{\text{rb}} = 1.000$.

E. LEAVE-ONE-COMPONENT-OUT ABLATION STUDY (RQ3)

To isolate the individual contribution of each architectural component, Table 6 presents a leave-one-component-out

ablation study across six representative instances covering clustered, random, and mixed topologies evaluated across 5 independent seeds (25 runs per instance).

1) Component Contribution Breakdown (RQ3)

- **Learned Acceptance Criterion (LAC):** On structured topologies such as $c2_2_1$, replacing LAC with static Simulated Annealing (Config 3) increases mean fleet size from 6.00 to 6.20 and degrades travel distance ($1931.44 \rightarrow 1944.40$), confirming that lookahead supervised labeling effectively prevents premature local stagnation on constrained clusters.



TABLE 6. Leave-One-Component-Out Ablation Study: Component Contribution Across 6 Representative Instances Covering Diverse Topologies and Scales (5 Independent Seeds).

Ablation Configuration	C101 (100-c)		R101 (100-c)		RC101 (100-c)		c2_2_1 (200-c)		r1_2_1 (200-c)		rc2_4_1 (400-c)	
	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD	NV	TD
BKS Baseline	10	828.94	19	1650.80	14	1696.95	6	1931.44	20	4784.11	11	6682.37
(1) Full Hybrid-DDQN	10.00	828.94	19.00	1653.12	15.20[†]	1677.28	6.00	1931.44	20.60[†]	5153.46	13.20[†]	6926.55
(2) w/o Macro Controller	10.00	828.94	19.00	1653.12	14.80 [†]	1700.21	6.00	1940.92	20.60 [†]	4985.55	12.80 [†]	6834.25
(3) w/o Learned Acceptance (LAC)	10.00	828.94	19.00	1653.12	15.40 [†]	1683.79	6.20 [†]	1944.40	20.20 [†]	4938.09	13.00 [†]	6798.10
(4) w/o HiGHS Recombination	10.00	828.94	19.00	1651.98	15.20 [†]	1688.95	6.00	1931.44	20.60 [†]	4885.57	12.80 [†]	6795.30
(5) w/o τ -Entropy Gate	10.00	828.94	19.00	1653.37	15.20 [†]	1668.40	6.00	1931.44	20.40 [†]	5001.86	13.00 [†]	6722.15
(6) Rule-Macro	10.00	828.94	19.00	1651.20	15.20 [†]	1664.73	6.00	1931.44	20.60 [†]	5300.45	13.00 [†]	6764.60
(7) Rule-Micro	10.00	828.94	19.00	1653.06	14.80 [†]	1693.75	6.60 [†]	2002.95	21.00 [†]	5160.67	13.60 [†]	6640.48

[†] Denotes instances with vehicle count exceeding BKS ($NV > NV_{BKS}$); travel distance comparisons against BKS must account for surplus vehicle capacity distortion.

- **HiGHS RoutePool Recombination:** Removing the Set Partitioning matheuristic (Config 4) degrades travel distance on mixed topologies such as RC101 (1677.28 $\rightarrow$ 1688.95, +11.67 km penalty), demonstrating the efficacy of recombining elite route fragments.
- **Scale-Aware Coordination Dynamics:** The full configuration provides a robust general-purpose compromise across diverse topologies. On extreme combinatorial scales ($N = 400$, $rc2_4_1$) and unconstrained random horizons ($r1_2_1$), multi-level coordination introduces exploration trade-offs, where simpler single-level ablations execute tighter greedy local exploitation.

VI. DISCUSSION AND MANAGERIAL INSIGHTS

This section interprets the empirical findings through the dual lenses of algorithmic properties and commercial fleet economics, contextualizing computational trade-offs, analytical properties, and limitations of learning-augmented metaheuristics.

A. SCALE-AWARE PERFORMANCE DYNAMICS: QUALITY-MAXIMIZATION VS. SEARCH SPEED

A central empirical finding emerging from our evaluation is that the performance advantage of Tri-Level Hybrid DDQN-ALNS exhibits a scale-aware divergence:

- **100- and 200-Customer Scale (NV-Floor Convergence & Distance Dominance):** On Solomon-100 and Homberger-200, both classical ALNS and Hybrid-DDQN reliably converge to the minimum vehicle count floor (e.g., $NV = 10.0$ on C1, $NV = 6.0$ on C2_200). In this regime, the hybrid advantage manifests as solution consistency and travel distance minimization: Hybrid-DDQN achieves travel distance savings across benchmark families, reducing the overall distance gap to BKS to +1.41% versus +3.73% for ALNS-Base. However, on complex mixed-distribution instances such as $rc1_2_5$, Hybrid-DDQN exhibits a noticeable +17.11% distance gap to BKS despite successfully matching the BKS vehicle count floor ($NV = 18.0$ vs. $NV = 19.0$ for ALNS-Base). This distance

inflation arises because strictly eliminating the 19th vehicle in mixed topologies forces the consolidation of dispersed customer clusters under tight delivery deadlines, resulting in elongated route detours that standard neighborhood operators cannot easily compact within standard iteration budgets without specialized cross-cluster trajectory bridges.

- **400-Customer Scale (Suboptimal Graceful Degradation):** Under extreme combinatorial dimensionality ($N = 400$), neither heuristic nor hybrid approaches attain BKS. However, Hybrid-DDQN demonstrates superior resistance to search stagnation, securing a vehicle reduction edge of 0.70 to 0.80 vehicles on clustered and random topologies ($c2_4_1$ and $r2_4_1$), while reducing travel distance by up to 10.66% (1309.67 km saved on $r1_4_1$).

Importantly, our framework is designed as a Quality-Maximizer rather than a search accelerator. While classical ALNS completes 2000 iterations in 1.0s to 15.0s due to raw heuristic evaluation, Hybrid-DDQN incurs an average runtime of 28.9s on Solomon-100, 58.6s on Homberger-200, and 160.9s on Homberger-400. This computational overhead stems from online neural forward/backward passes, prioritized replay buffer management, and periodic exact Set Partitioning solves via HiGHS. On small, lightly constrained instances ($N \leq 100$), mature domain-specific genetic algorithms (such as HGS-VRPTW) achieve marginal distance advantages (+0.13% vs. +1.41%) through fast generational crossover; however, Hybrid-DDQN's primary strength lies in maintaining strict operational constraint feasibility, preventing search stagnation, and consistently driving fleet reductions on large, highly constrained topologies ($N = 200, 400$).

1) Learned Control vs. Handcrafted Rule Thresholds

To test whether learned, state-conditioned control is necessary or whether a hand-tuned deterministic policy suffices, Table 6 (configs 6–7) introduces **Rule-Macro** (deterministic five-branch tree replacing π_{macro} over the same 12-dimensional state) and **Rule-Micro** (analogous tree replacing π_{micro} over the 20-dimensional state), each isolated

while LAC and HiGHS recombination remain unchanged. On *RC101*, Rule-Micro obtains a lower mean fleet size ($NV = 14.80$) than Full Hybrid-DDQN ($NV = 15.20$), the primary lexicographic metric, though at a higher mean travel distance (1693.75 vs. 1677.28). On *r1_2_1*, Full Hybrid-DDQN matches Rule-Macro on mean NV (20.60) while achieving lower mean travel distance (5153.46 vs. 5300.45), and outperforms Rule-Micro on both metrics ($NV = 20.60$ vs. 21.00). On *R101*, the three configurations converge to nearly identical outcomes ($NV = 19.00$ across all three; TD within 0.2% of one another). Rule-Macro also improves upon Full Hybrid-DDQN on *rc2_4_1*, in both mean fleet size ($NV = 13.00$ vs. 13.20) and mean travel distance (6764.60 vs. 6926.55). Taken together, these observations demonstrate that while handcrafted decision trees can be calibrated to excel on specific instance subclasses, state-conditioned Dueling DDQN provides adaptive, general-purpose robustness across the full spectrum of spatial-temporal distributions without requiring manual threshold re-engineering.

B. MANAGERIAL IMPLICATIONS FOR COMMERCIAL LOGISTICS

In commercial freight transportation and last-mile parcel distribution, fleet operational expenses are heavily driven by capital expenditures, vehicle licensing, insurance, and driver shift wages rather than marginal fuel consumption [?], [?]. Fixed operational overheads typically represent the dominant share of operating budgets, while fuel expenditure contributes a secondary portion.

Consequently, eliminating a single vehicle in a commercial dispatch depot generates substantial recurring cost savings compared to marginal distance reductions. In pre-dispatch planning workflows where routes are generated overnight or during 15- to 30-minute planning windows prior to morning delivery waves, allocating additional computational time to secure vehicle fleet reductions and travel distance savings represents a favorable operational trade-off.

C. ANALYTICAL PROPERTIES AND COMPLEXITY CONSIDERATIONS

To substantiate the empirical behavior, we examine four analytical properties characterizing the algorithmic mechanics.

1) Property 1 (Lexicographic Objective Consistency)

Let $s_1, s_2 \in \mathcal{S}$ be feasible routing solutions. Under the scalar surrogate objective $F(s) = \mathcal{M} \cdot N_V(s) + D_T(s)$, if the penalty weight satisfies $\mathcal{M} > \max_{s \in \mathcal{S}} D_T(s) - \min_{s \in \mathcal{S}} D_T(s)$, then $F(s_1) < F(s_2) \iff s_1 \prec s_2$. In our implementation, $\mathcal{M} = 10^5$ was selected above the maximal total route length across all evaluated benchmark instances, guaranteeing lexicographical equivalence.

2) Property 2 (Granular Candidate Space Reduction)

For a VRPTW instance with N customers, dynamic active fleet size $K(s)$, and top- k spatiotemporal candidate filtering, the theoretical pairwise candidate reduction ratio Γ_{prune} for

1-edge insertion moves relative to exhaustive pairwise evaluation is given by:

$$\Gamma_{\text{prune}}(N, K, k) = 1 - \frac{2kK(s)}{N(N-1)} \quad (30)$$

where $K(s)$ is the number of active routes at state s . For $N = 200, K = 10, k = 25$, this yields an evaluation space reduction of $\Gamma_{\text{prune}} = 98.74\%$, and $\Gamma_{\text{prune}} = 99.37\%$ on $N = 400, K = 20, k = 25$. This bound holds strictly for local single-edge substitutions; for multi-customer cluster operations (e.g., Shaw removal), repair complexity is further bounded by candidate adjacency within the surviving route graph.

3) Property 3 (Multi-Scale Set Partitioning Penalty Grid)

In Set Partitioning recombination over Route Pool $\mathcal{P}$, the multi-scale penalty grid $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\}\bar{c}$ provides a systematic progression across exploration regimes, ranging from distance preservation to aggressive vehicle elimination, preventing numerical stalling during exact MIP solves.

4) Property 4 (Downstream Lookahead under Learned Acceptance)

In classical Simulated Annealing, the acceptance probability for a non-improving move $\Delta C > 0$ decays exponentially as temperature cools ($P_{\text{SA}} = \exp(-\Delta C/T) \rightarrow 0$ as $T \rightarrow 0$), restricting late-stage search to strictly downhill transitions. In contrast, the Learned Acceptance Criterion augments Simulated Annealing by evaluating structural transition promise based on delayed lookahead labeling ($H_{\text{lac}} = 50$). When the classifier predicts a high probability of future incumbent discovery ($p_t \geq \tau_{\text{acc}}$), non-improving intermediate moves are retained, facilitating plateau escape during deep cooling stages.

D. LIMITATIONS AND THREATS TO VALIDITY

While Tri-Level Hybrid DDQN-ALNS exhibits strong empirical performance, five primary limitations should be noted:

- 1) **Computational Latency on Small Instances:** On small-scale problems ($N \leq 50$), classical heuristics find near-optimal solutions in fractions of a second. The online neural training and replay buffer maintenance of Hybrid-DDQN introduce unnecessary overhead in this regime without substantial solution quality gains.
- 2) **MIP Solver Dependency:** The performance of the matheuristic recombination tier depends on the availability and efficiency of an underlying MIP solver (HiGHS). On dense route pools ($|\mathcal{P}| > 2000$), solve time limits (4.0s) must be enforced to prevent solver stalling.
- 3) **Late-Stage Lookahead Horizon Truncation in LAC:** The LAC's 50-iteration lookahead horizon is calibrated for local plateau escapes and becomes increasingly conservative in late-stage search as class imbalance

($y_t = 1$ rate $< 2\%$) biases the online MLP toward low acceptance probabilities. Compound fleet-reduction restructuring requiring > 50 iterations is explicitly delegated to the Macro Controller's HiGHS and thermal reheating modes.

- 4) **Fixed-Iteration vs. Equal Wall-Clock Comparison:** Our primary benchmark suite evaluated algorithms under identical iteration limits ($T_{\max} = 2000$). Because learning-augmented components introduce per-step computational overhead, Hybrid-DDQN utilizes a larger wall-clock budget than baseline ALNS.
- 5) **Online Adaptation Scope:** The framework relies on online instance-adaptive reinforcement learning during the search trajectory. For sub-second dynamic dispatch scenarios with extreme real-time network distortions, offline pretrained policy initialization may be required.

VII. CONCLUSION

This paper presented Tri-Level Hybrid DDQN-ALNS, a learning-augmented optimization framework for the Vehicle Routing Problem with Time Windows (VRPTW). By decomposing metaheuristic search management into a hierarchical decision structure, our method addresses key limitations of both classical stateless heuristics and pure end-to-end neural solvers:

- 1) **Hierarchical Decision Architecture:** Decoupling macro-level search exploration regimes (Plateau Controller) from micro-level destroy-repair operator selection (Dueling DDQN with Softmax-Entropy confidence gating) and learned acceptance control (LAC) enables coordinated plateau escape while preserving the auditability of neighborhood operators.
- 2) **Strict Feasibility & Lexicographic Alignment:** Hard temporal and capacity constraints are strictly enforced through $O(1)$ Savelsbergh forward time slack checks and granular spatiotemporal filtering, while multi-objective rewards and exact Set Partitioning recombination over Route Pool $\mathcal{P}$ preserve the primary-secondary fleet minimization hierarchy ($N_V \succ D_T$).
- 3) **Comprehensive Empirical Validation:** Extensive benchmark evaluations across 74 benchmark instances (Solomon-100, Homberger-200, and Homberger-400) under strictly isolated independent cold-starts demonstrate that our method significantly reduces fleet size ($p = 6.91 \times 10^{-7}$) and achieves a 3.73% travel distance reduction under matched fleet counts with zero losses ($p = 9.31 \times 10^{-8}$), passing Bonferroni-corrected significance thresholds.

A. TRADE-OFFS AND LIMITATIONS

Our empirical findings also delineate the operational scope of the method. The framework is designed as a quality-maximizing optimization pipeline that trades additional computational time for enhanced solution quality, rather than a sub-second search accelerator. On small-scale instances ($N \leq 50$), classical heuristics remain computationally

preferable. Furthermore, the recombination tier relies on an exact MIP solver, and performance on extreme out-of-distribution topologies depends on policy calibration.

B. FUTURE RESEARCH DIRECTIONS

Several promising research directions emerge from this work:

- **Electric Vehicle Routing (EV-VRPTW):** Extending the state space and operator portfolio to incorporate non-linear battery charging profiles, energy recovery, and charging station queuing dynamics.
- **Heterogeneous Multi-Depot Operations:** Generalizing the Macro Plateau Controller to manage inter-depot vehicle transfers and multi-depot fleet capacity constraints.
- **Dynamic Real-Time Re-Optimization:** Adapting the lightweight neural operator selection policy for sub-second reactive re-routing under stochastic customer demand and real-time traffic congestion.
- **Suite-Wide Empirical Ablations:** Scaling component leave-one-out and heuristic-rule sensitivity ablations across the full 74-instance testbed to conduct formal multi-factor ANOVA across diverse spatial-temporal topologies.



HUYNH NHAT HUY is currently pursuing the B.S. degree in Computer Science with the Faculty of Information Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam. His research interests include deep reinforcement learning, combinatorial optimization, metaheuristics, and learning-augmented algorithms.



THI-LINH HO received her B.S. degree in information technology from Nong Lam University, Ho Chi Minh City, Vietnam in 2009. She received her master degree in Management Information System in 2012 from Ho Chi Minh City University of Technology, Vietnam. She then received her doctorate in computer science from Ton Duc Thang University, Vietnam in 2024. She is a lecturer at the Faculty of Information Technology, Ton Duc Thang University. She is also a member of the Natural Language Processing and Knowledge Discovery Research Group. Her fields of interest consist of working with machine learning, intelligent information systems, analyzing data and solving optimization problems in business, finance, economics, etc.

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